

What is it ?

- A data structure of nodes (vertices) and their connections (edges)
- All trees are graphs, not all graphs are trees

Terminology of Graphs

- cycle: When you start at Node(x), follow the links, and end back at Node(x)
- acyclic: A graph that contains no cycles
- connected: When every node has a path to another node
- directed: When there is a direction to the connections. Think Twitter
- undirected: !directed. Think Facebook
- weighted: The edges have a weight associated with them. Think Maps
- DAG: Directed, acyclic graph.

How are graphs represented ?

- adjacency list
- adjacency matrix